

Question 3 (Begin a new page)

- a) (i) Show that the x coordinates of the stationary points on the graph $y = \{f(x)\}^2$ are the same as the x coordinates of the stationary points or intercepts on the x -axis of the graph $y = f(x)$.
- (ii) $f(x) = \sin^2 x - \frac{1}{2}$, $0 \leq x \leq \pi$. On the same axes, and without using further calculus, sketch the graphs of $y = f(x)$ and $y = \{f(x)\}^2$.
- b) (i) Sketch the graph of $y = x^2 + \frac{2}{x}$ showing any intercepts on the coordinate axes, asymptotes, stationary points or points of inflexion.
- (ii) The equation $x^2 + \frac{2}{x} - k = 0$ has exactly two different real solutions. Find the value of k and the real solutions of the equation.
- c) (i) Find $\int \sqrt{e^x} dx$
- (ii) Find $\int (\cos^2 x - \sin^2 x) dx$

Question 4 (Begin a new page)

- a) Consider the function $f(x) = \frac{3x}{(x-1)(4-x)}$.
- (i) Express $f(x)$ in partial fractions.
- (ii) Find the coordinates and nature of any turning points on the graph $y = f(x)$.
- (iii) Sketch the graph $y = f(x)$ showing clearly the coordinates of any turning points and the equations of any asymptotes.
- (iv) Find the area of the region bounded by the curve $y = f(x)$ and the x -axis between the lines $x = 2$ and $x = 3$.

3.

b) Consider the function $f(x) = \frac{x-1}{x}$.

- (i) Sketch the graph $y = f(x)$ showing clearly the coordinates of any points of intersection with the axes and the equations of any asymptotes.
- (ii) Use the graph $y = f(x)$ to sketch on separate axes the graphs
 - (α) $y = |f(x)|$
 - (β) $y = f(|x|)$
 - (γ) $y = \{f(x)\}^2$

Question 5 (Begin a new page)

a) (i) Show that the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(a \cos \theta, b \sin \theta)$ in the first quadrant has equation $bx \cos \theta + ay \sin \theta - ab = 0$.

(ii) This tangent cuts the x-axis at point A and the y-axis at point B. Find the minimum area of $\triangle AOB$ and show that when this occurs P is the midpoint of AB.

b) (i) Show that the tangent to the rectangular hyperbola $xy = 4$ at the point $T(2t, \frac{2}{t})$ has equation $x + t^2 y = 4t$.

(ii) This tangent cuts the x-axis at point Q. Show that the line through Q which is perpendicular to the tangent at T has equation $t^2 x - y = 4t^3$.

(iii) The line through Q cuts the rectangular hyperbola at the points R and S. Show that the midpoint M of RS has coordinates $M(2t, -2t^3)$.

Question 6 (Begin a new page)

a) Find $\int \frac{x+1}{\sqrt{x-1}} dx$ using the substitution $u = x-1$

b) Evaluate $\int_1^2 \frac{1}{e^x - 1} dx$

c) Evaluate $\int_0^{\pi/2} \frac{1}{\cos x + 2} dx$ using the substitution $t = \tan \frac{x}{2}$.

4.

d) (i) If $I_n = \int_0^{\pi/4} \tan^n x \, dx$, $n = 0, 1, 2, \dots$

show that $I_n + I_{n-2} = \frac{1}{n-1}$, $n = 2, 3, 4, \dots$

(ii) Hence find the value of $\int_0^{\pi/4} \tan^5 x \, dx$.

Question 7 (Begin a new page)

a) The base of a particular solid is the region bounded by the parabola $y^2 = 4x$ between its vertex $(0,0)$ and its latus rectum. Find the volume of the solid if every cross section perpendicular to the x-axis

- (i) is a square with one side in the base of the solid
- (ii) is an equilateral triangle with one side in the base of the solid.

b) The equation $x^3 + kx + 2 = 0$ has roots α, β and γ .

- (i) Find the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.
- (ii) Find the monic equation with roots α^2, β^2 and γ^2 .
- (iii) Show that the value of $\alpha^3 + \beta^3 + \gamma^3$ is independent of k .

Question 8 (Begin anew page)

a) An ellipse has equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$.

- (i) Find the ellipses eccentricity.
- (ii) Find the coordinates of the foci S and S^1 and the equations of the directrices.
- (iii) Sketch the ellipse showing the above features.
- (iv) If $P(x_1, y_1)$ is a point on the ellipse show that $PS + PS^1$ is independent of the position of P .

5.

- b) (i) Sketch the graph of $y = x^2 - 4$ showing the intercepts on the axes.
- (ii) Without using calculus, use the graph of $y = x^2 - 4$ to sketch on separate axes the graphs of
- (α) $y = |x^2 - 4|$
- (β) $y = \frac{1}{x^2 - 4}$
- (γ) $y^2 = x^2 - 4$
- In each case show clearly any intercepts on the axes and any asymptotes.

Question 1 (17)

a $\int \frac{x}{\sqrt{x-1}} dx$

let $u = x-1$
 $\therefore du = dx$

$= \int \frac{u+1}{\sqrt{u}} du$

and $x = u+1$

$= \int \frac{u}{u^{\frac{1}{2}}} + \frac{1}{u^{\frac{1}{2}}} du$

(84)

$= \int (u^{\frac{1}{2}} + u^{-\frac{1}{2}}) du$

$= \frac{2}{3} u^{\frac{3}{2}} + 2u^{\frac{1}{2}} + C$ or $\frac{2}{3} u^{\frac{1}{2}} (u+3) + C$

$= \frac{2}{3} \sqrt{(x-1)^3} + 2\sqrt{x-1} + C = \frac{2}{3} \sqrt{x-1} (u+2) + C$

b $\int_{\pi/4}^{\pi/3} (1 + \tan^2 x) \tan x dx$

$\tan^2 x + 1 = \sec^2 x$

$= \int_{\pi/4}^{\pi/3} \sec^2 x \tan x dx$ ✓

$= \left[\frac{1}{2} \tan^2 x \right]_{\pi/4}^{\pi/3}$ ✓

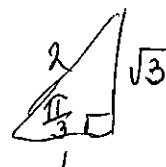
$= \frac{1}{2} \tan^2 \frac{\pi}{3} - \frac{1}{2} \tan^2 \frac{\pi}{4}$

$= \frac{1}{2} \cdot \sqrt{3}^2 - \frac{1}{2} \cdot 1$

$= \frac{1}{2} \cdot 3 - \frac{1}{2}$ ✓

$= 1$

(3)



Question 1

$$\tan^2 + 1 = \sec^2$$

b ii $\int_0^{\ln 2} e^x \sec^2(e^x) dx.$

$$= \left[\tan(e^x) \right]_0^{\ln 2} \quad \checkmark$$

$$= \tan e^{\ln 2} - \tan 1$$

$$= \tan 2 - \tan 1. \quad (\approx 3.74) \quad \checkmark$$

(2)

c $\int_0^{\pi/2} \frac{1}{7+5\sin x + 5\cos x} dx$

$$= \int_0^1 \left(\frac{1}{7+5\left(\frac{2t}{1+t^2}\right)+5\left(\frac{1-t^2}{1+t^2}\right)} \right) \cdot \frac{2dt}{(1+t^2)} \quad \checkmark$$

$$= \int_0^1 \frac{2dt}{7(1+t^2) + 10t + 5 - 5t^2}$$

$$= \int_0^1 \frac{2dt}{7+7t^2+10t+5-5t^2}$$

$$= \int_0^1 \frac{2dt}{2t^2+10t+12}$$

$$= \int_0^1 \frac{dt}{t^2+5t+6} \quad \checkmark$$

$$= \int_0^1 \frac{dt}{(t+3)(t+2)}$$

$$= \int_0^1 \frac{1}{t+2} - \frac{1}{t+3} dt.$$

$$= \left[\ln(t+2) - \ln(t+3) \right]_0^1 = \ln \frac{9}{8} \quad \checkmark$$

let $t = \tan \frac{x}{2}$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx.$$

$$= \frac{1}{2} (\tan^2 \frac{x}{2} + 1) dx$$

$$= \frac{1}{2} (t^2 + 1) dx.$$

$$\frac{2dt}{t^2+1} = dx \quad \checkmark$$

and $\sin x = \frac{2t}{1+t^2}$

$$\cos x = \frac{1-t^2}{1+t^2}$$

when $x=0 \Rightarrow t=0$

$$x = \frac{\pi}{2} \Rightarrow t=1.$$

(4)

Question 1

$$\frac{d}{dx} \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\ln x}{1+x^2} dx.$$

$$= \int_{\sqrt{3}}^{\frac{\sqrt{3}}{1}} \frac{\ln(\frac{1}{u})}{1+\frac{1}{u^2}} \cdot -\frac{1}{u^2} du. \checkmark$$

$$= \int_{\sqrt{3}}^{\frac{\sqrt{3}}{1}} \frac{\ln 1 - \ln u}{u^2 + 1} \cdot -1 du.$$

$$= \int_{\sqrt{3}}^{\frac{\sqrt{3}}{1}} \frac{\ln u}{1+u^2} du$$

$$= - \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\ln u}{1+u^2} du.$$

$$= - \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\ln x}{1+x^2} dx. \checkmark$$

$$\therefore \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\ln x}{1+x^2} dx = - \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\ln x}{1+x^2} dx$$

$$2 \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\ln x}{1+x^2} dx = 0.$$

$$\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\ln x}{1+x^2} dx = 0 \checkmark$$

$$u = \frac{1}{x} = x^{-1}$$

$$du = -x^{-2} dx.$$

$$= -\frac{1}{x^2} dx.$$

$$= -u^2 dx$$

$$-\frac{1}{u^2} du = dx$$

$$\text{when } x = \sqrt{3} \Rightarrow u = \frac{\sqrt{3}}{3}$$

$$\text{when } x = \frac{1}{\sqrt{3}} \Rightarrow u = \sqrt{3}$$

$$x^2 = \frac{1}{u^2}$$

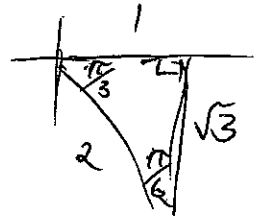
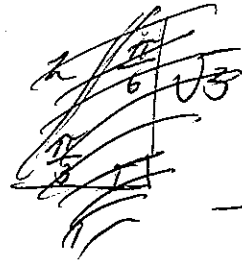
(7)

Question 2 (15)

a i $z = 2 - 2\sqrt{3}i$

$$= 4\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$$

$$= 4\left\{\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right\}$$



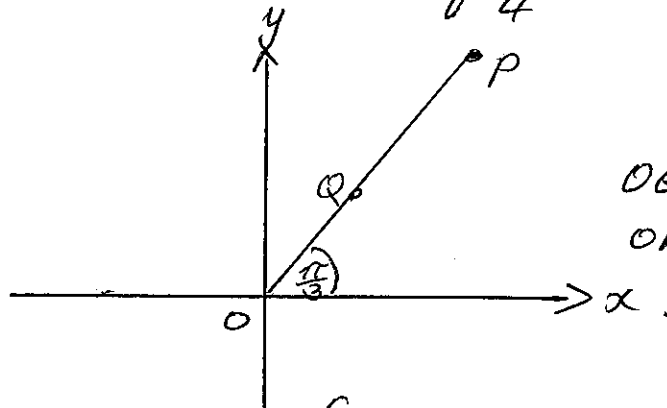
or

ii

$\hat{z}$ has modulus of 4 and argument of $\frac{\pi}{3}$

$\frac{1}{z}$ has modulus of $\frac{1}{4}$ and argument of $-\frac{\pi}{3}$

(2)



$$OQ = \frac{1}{4}$$

$$OP = 4.$$

OQ represents $\frac{1}{z}$
OP represents $\hat{z}$.

b i) let $z = x + iy$. $x \geq 0$. ✓

$$2|z| = 3(z + \bar{z})$$

$$4|z|^2 = 9(z + \bar{z})^2$$

$$4(x^2 + y^2) = 9(2x)^2$$

$$4x^2 + 4y^2 = 36x^2$$

$$4y^2 = 32x^2$$

$$y^2 = 8x^2 \therefore y = \pm 2\sqrt{2}x$$

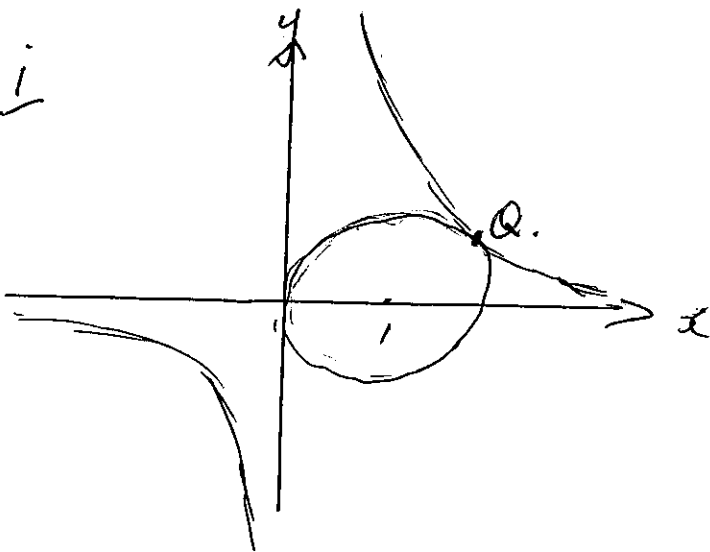
(4)

Question 2

b ii

①.

c i



①

ii if β is a ^{repeated} root of $P(x) = 0$ then there is a polynomial $Q(x)$ such that

$$P(x) = (x - \beta)^2 Q(x). \quad \checkmark$$

$$\begin{aligned} \therefore P'(x) &= 2(x - \beta)Q(x) + Q'(x)(x - \beta)^2 \\ &\checkmark = (x - \beta) [2Q(x) + (x - \beta)Q'(x)] \end{aligned}$$

$$\therefore P'(\beta) = 0. \quad \checkmark$$

Question 2

iii At Q in i

$$xy = c^2$$

$$y = \frac{c^2}{x}$$

$$y^2 = \frac{c^4}{x^2} \text{ --- (1) and } (x-1)^2 + y^2 = 1 \text{ --- (2)}$$

substitute (1) into (2). ✓

$$(x-1)^2 + \frac{c^4}{x^2} = 1.$$

$$x^2(x-1)^2 + c^4 = x^2$$

Now, since the circle and hyperbola touch at Q, then the x-coordinate of Q is a repeated real root of this equation.

There are no other real roots since the circle and hyperbola have no other intersection points.

The remaining 2 roots are complex roots that are complex conjugates (α and $\bar{\alpha}$) since the equation has real coefficients.

Question 3 (14)

a i

$$y = \{f(x)\}^2$$

$$\frac{dy}{dx} = 2f(x) \cdot f'(x) \quad (1)$$

$$\frac{dy}{dx} = 0 \Rightarrow f^2(x) = 0 \text{ or } f'(x) = 0$$

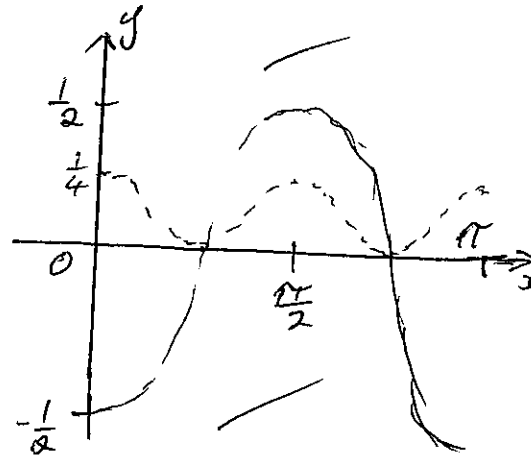
ii $f(x) = \sin^2 x - \frac{1}{2}$

when $f(x) = 0$

then $\sin^2 x = \frac{1}{2}$ ✓

$$x = \frac{\pi}{4}, \frac{3\pi}{4}$$

(3)



b i

$$y = x^2 + \frac{2}{x}$$

* as $x \rightarrow 0$, $y \rightarrow \infty$ $x \neq 0$
 $\therefore y$ is an asymptote.

* $y = 0 \quad \therefore x^2 = -\frac{2}{x}$
 $x^3 = -2$

* $\frac{dy}{dx} = 2x - \frac{2}{x^2}$

$$\frac{dy}{dx} = 0 \Rightarrow \frac{2}{x^2} = 2x$$

$$x^3 = 1$$

$$x = 1$$

* $\frac{d^2y}{dx^2} = 2 + \frac{4}{x^3}$

$$\frac{d^2y}{dx^2} = 0 \Rightarrow \frac{4}{x^3} = -2$$

$$x^3 = -2$$

$$x = -\sqrt[3]{2}$$

* when $x = 1$, $\frac{d^2y}{dx^2}$ is (+)ve
 $\therefore$ Min. Point at $(1, 3)$.

* when $x = -\sqrt[3]{2}$, $\frac{d^2y}{dx^2} =$

x	$< -\sqrt[3]{2}$	$>$
y''	$+$	$-$

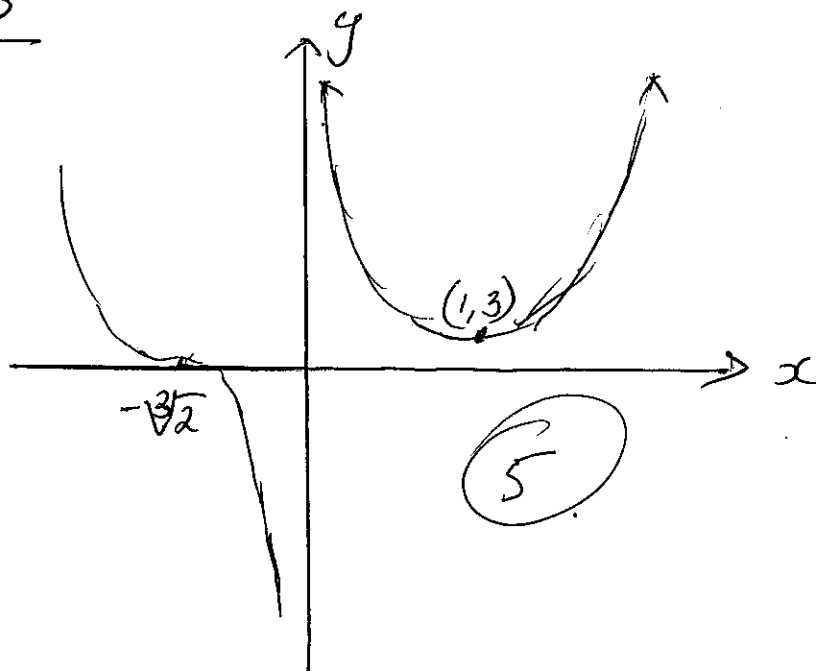
$\therefore$ Point of T. Maximum at $(-\sqrt[3]{2}, 0)$.

Question 3

b i

* as $x \rightarrow \infty$,
 $y \rightarrow \infty$.

* as $x \rightarrow -\infty$
 $y \rightarrow \infty$



b ii $y = x^2 + \frac{2}{x} - b$ is the same as the above graph moved down 'b' units. For $x^2 + \frac{2}{x} - b = 0$ to have exactly 2 real roots (different), the graph touches the x-axis at $x = 1$.
 $\therefore$ the graph is moved down 3 units.

$$\therefore b = 3$$

$$\therefore x^2 + \frac{2}{x} - 3 = 0$$

$$x^3 - 3x + 2 = 0.$$

$x = 1, 1$ are the 2 equal roots

By inspection, the other root is -2.

$$(x-1)(x-1)(x+2) = 0.$$

$\therefore$ the 2 real solutions are 1 and -2.

3

Question 3

$$\begin{aligned} \underline{\text{c i}} \quad \int \sqrt{e^x} \, dx &= \int e^{\frac{x}{2}} \, dx. \\ &= 2e^{\frac{x}{2}} + c \quad \checkmark \\ &= 2\sqrt{e^x} + c. \quad (1) \end{aligned}$$

$$\begin{aligned} \underline{\text{ii}} \quad \int (\cos^2 x - \sin^2 x) \, dx &= \int \cos 2x \, dx. \\ &= \frac{1}{2} \sin 2x + c \quad \checkmark \\ &\quad (1) \end{aligned}$$

Question 4 (16)

$\frac{a}{1}$

$$f(x) = \frac{3x}{(x-1)(4-x)}$$

$$\text{let } \frac{3x}{(x-1)(4-x)} = \frac{A}{x-1} + \frac{B}{4-x}$$

$$\checkmark = \frac{A(4-x) + B(x-1)}{(x-1)(4-x)}$$

Equating numerators

$$3x = A(4-x) + B(x-1)$$

$$\text{when } x=4$$

$$12 = 3B$$

$$B = 4$$

$$\text{when } x=1$$

$$3 = 3A$$

$$A = 1 \checkmark$$

(2)

$$\therefore f(x) = \frac{1}{x-1} + \frac{4}{4-x}$$

ii

$$f(x) = \frac{1}{x-1} + \frac{4}{4-x}$$

$$= (x-1)^{-1} + 4(4-x)^{-1}$$

$$f'(x) = -1(x-1)^{-2} + 4(4-x)^{-2}$$

$$= \frac{-1}{(x-1)^2} + \frac{4}{(4-x)^2}$$

$$= \frac{-(4-x)^2 + 4(x-1)^2}{(x-1)^2(4-x)^2}$$

$$f'(x) = 0 \quad \text{when } 4(x-1)^2 = (4-x)^2$$

$$\text{ie when } x = \pm 2.$$

✓✓

$$f''(x) = 2(x-1)^{-3} + 8(4-x)^{-3}$$

(3)

$$\text{when } x=2 (y=3)$$

$\therefore f''(2)$ is positive

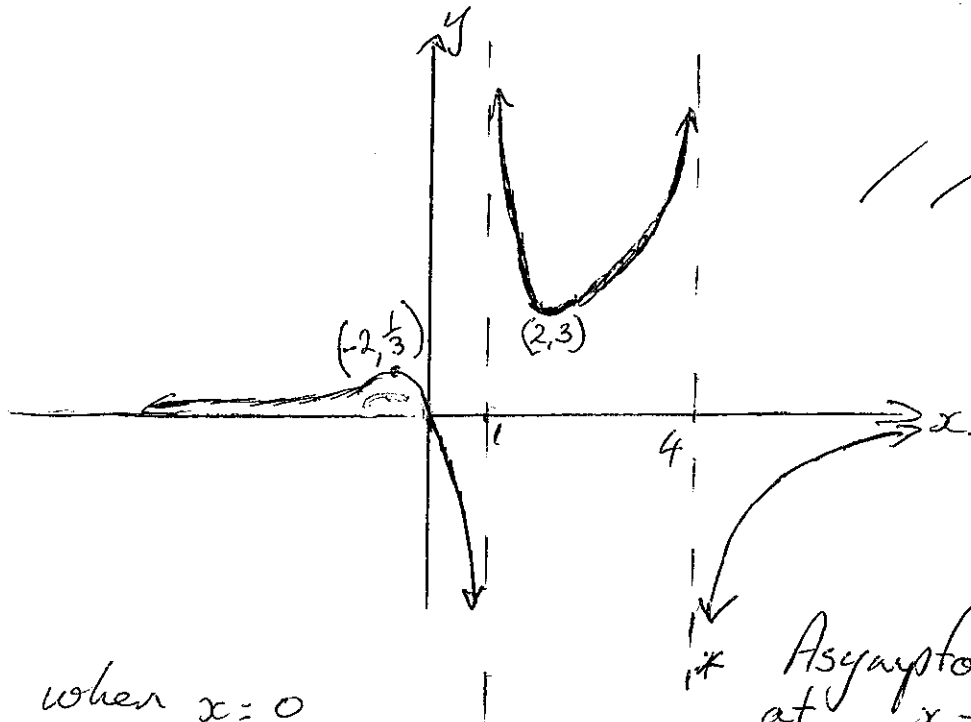
$\therefore$ Min T. Point at $(2, 3)$

$$\text{when } x=-2 (y=\frac{1}{3})$$

$f''(-2) < 0 \therefore$ Max T. Point $(-2, \frac{1}{3})$

Q
iii

Question 4



* when $x=0$
 $f(x)=0$

* Asymptotes
at $x=1$
and $x=4$

IV Area $= \int_a^b f(x) dx$

$$= \int_2^3 \frac{1}{x-1} + \frac{4}{4-x} dx$$

$$= \left[\ln(x-1) - 4 \ln(4-x) \right]_2^3$$

$$= (\ln 2 - 4 \ln 1) - (\ln 1 - 4 \ln 2)$$

$$= \ln 2 + 4 \ln 2$$

$$= 5 \ln 2 \text{ units}^2$$

* as $x \rightarrow +\infty$
 $f(x) \rightarrow -0$

* as $x \rightarrow +4$
 $f(x) \rightarrow -\infty$

* as $x \rightarrow -\infty$
 $f(x) \rightarrow +0$

2

Question 4.

b
i

$$f(x) = \frac{x-1}{x} \left(1 - \frac{1}{x}\right)$$

* $x \neq 0$
 $\therefore$ asymptote at $x=0$.

* when $x=1$, $f(x) = 0$.

$$\begin{aligned} * f'(x) &= \frac{x - (x-1)}{x^2} \\ &= \frac{1}{x^2} (x^{-2}) \end{aligned}$$

$$\begin{aligned} f''(x) &= -2x^{-3} \\ &= -\frac{2}{x^3} \end{aligned}$$

* $f'(x) \neq 0$.
 $\therefore$ there are no turning points

$$* \text{ let } y = \frac{x-1}{x}$$

$$xy = x-1$$

$$xy - x = -1$$

$$x - xy = 1$$

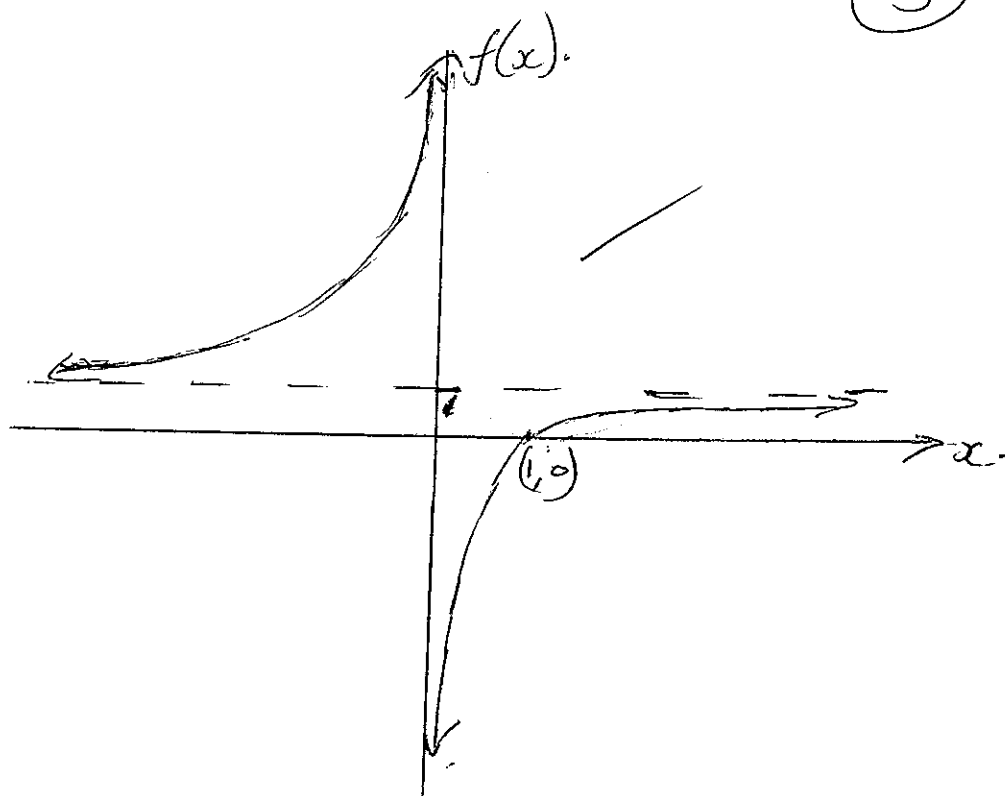
$$x(1-y) = 1$$

$$x = \frac{1}{1-y}$$

$$y \neq 1.$$

$\therefore$ there is a horizontal asymptote at $y=1$.

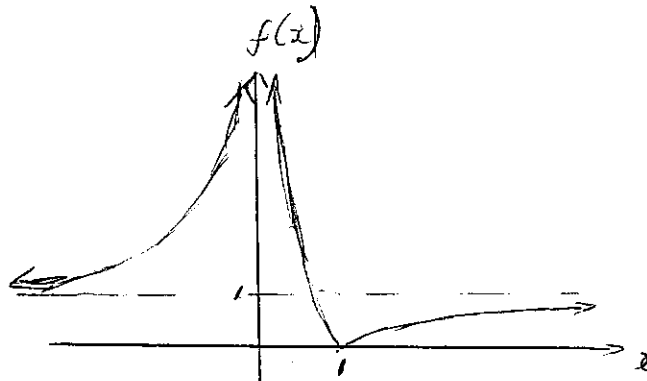
(3)



Question 4

b ii

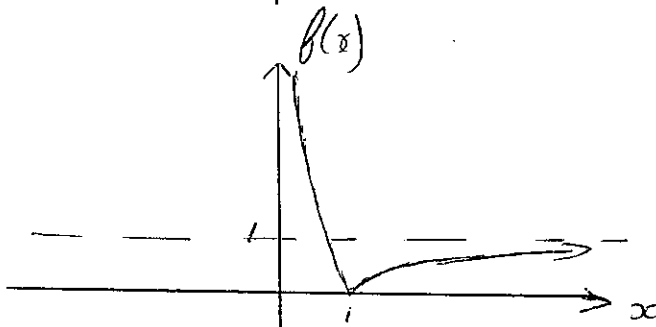
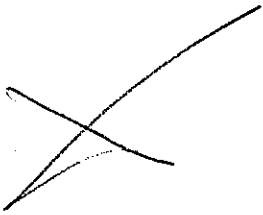
(∞)



$$|f(x)|$$

(7)

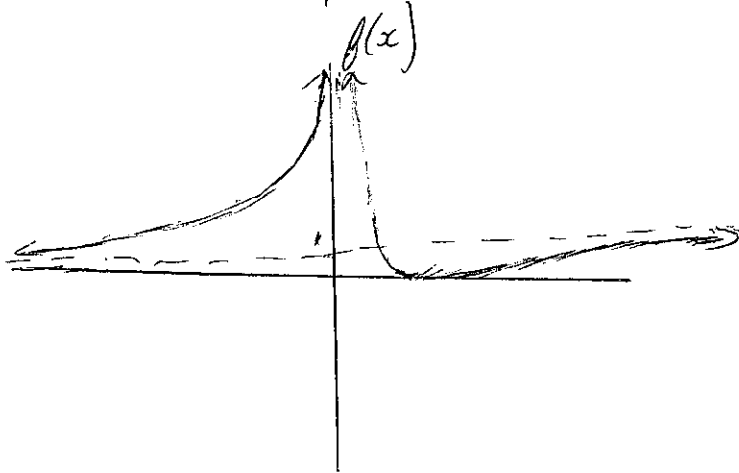
(β)



$$\beta(|x|)$$

(1)

(α)



$$y = \{\beta(x)\}^2$$

(1)

Question 5. (14)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{2x}{a^2} \cdot \frac{b^2}{2y}$$

$$= -\frac{b^2}{a^2} \cdot \frac{x}{y}$$

$$\begin{aligned} \text{at } P, \text{ grad of tangent } m &= -\frac{b^2}{a^2} \times \frac{a \cos \theta}{b \sin \theta} \\ &= -\frac{b \cos \theta}{a \sin \theta} \end{aligned}$$

(3) Equation of Tangent at $P(a \cos \theta, b \sin \theta)$

$$y - y_1 = m(x - x_1)$$

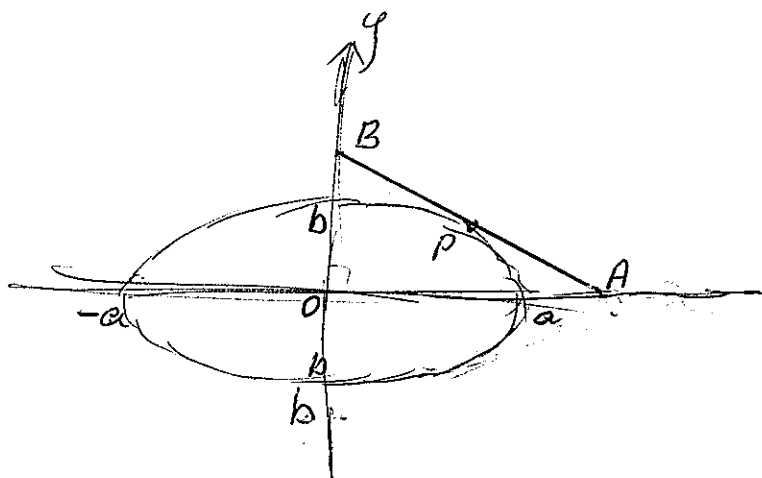
$$y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$a y \sin \theta - a b \sin^2 \theta = -b x \cos \theta + a b \cos^2 \theta$$

$$b x \cos \theta + a y \sin \theta - a b (\sin^2 \theta + \cos^2 \theta) = 0$$

$$b x \cos \theta + a y \sin \theta - a b = 0 \quad \text{--- (1)}$$

ii



Question 5

Q ii From (1) when $x=0$

$$ay \sin \theta - ab = 0$$

$$y = \frac{ab}{a \sin \theta}.$$

$$y = \frac{b}{\sin \theta}.$$

$\therefore$ B has coordinates $(0, \frac{b}{\sin \theta})$ or $(0, b \csc \theta)$

From (1) when $y=0$

$$bx \cos \theta - ab = 0$$

$$x = \frac{ab}{b \cos \theta}.$$

$$x = \frac{a}{\cos \theta}.$$

$\therefore$ A has coordinates $(\frac{a}{\cos \theta}, 0)$ or $(a \sec \theta, 0)$.

$$\text{Area of } \triangle AOB = \frac{1}{2} \times b \times h.$$

$$= \frac{1}{2} \times \frac{a}{\cos \theta} \times \frac{b}{\sin \theta}.$$

$$= \frac{ab}{2 \sin \theta \cos \theta}.$$

~~Wojan~~

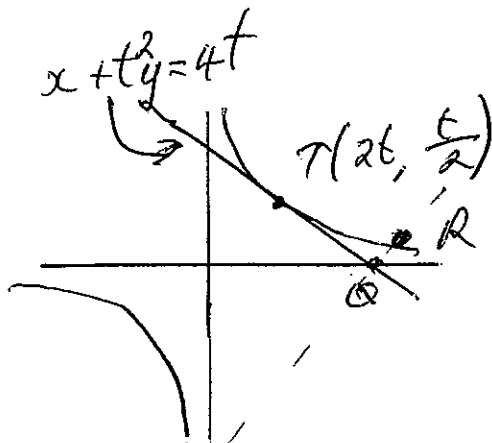
Question 5

b i

Calculus

(3)

ii



for $x + t^2 y = 4t$

when $y = 0$, $x = 4t$.

$\therefore Q$ is $(4t, 0)$

gradient of tangent is $-\frac{1}{t^2}$

$\therefore$ gradient of line through Q is t^2 .

Eqn of this line is

$$y = t^2(x - 4t)$$

$$y = t^2 x - 4t^3$$

$$t^2 x - y = 4t^3$$

iii

solving $t^2 x - y = 4t^3$ and $xy = 4$
substituting gives $y = \frac{4}{x}$

$$t^2 x^2 - 4xt^3 - 4 = 0.$$

Question 5

6 iii Continued.

Now, this is a quadratic

The sum of the roots is; $\alpha + \beta = -\frac{b}{a}$
 $= -\frac{-4t^3}{t^2}$
 $= 4t$

Let R and S have x values α and β .

The midpoint of R and S is given by $\frac{\alpha + \beta}{2}$ (x-values).

$$\therefore \frac{\alpha + \beta}{2} = \frac{4t}{2} = 2t \quad (\text{x-value}).$$

substitute into $t^2x - y = 4t^3$

$$\text{gives } 2t^3 - y = 4t^3$$

$$y = -2t^3$$

$$\therefore H \text{ is } (2t, -2t^3).$$

✓ (3)

Question 6 (15)

a $\int \frac{x+1}{\sqrt{x-1}} dx.$

$$\begin{aligned} u &= x-1 \\ du &= dx. \\ x &= u+1 \end{aligned}$$

$$= \int \frac{u+1+1}{\sqrt{u}} du \quad \checkmark$$

$$= \int \frac{u+2}{\sqrt{u}} du$$

$$= \int u^{1/2} + 2u^{-1/2} du \quad \checkmark$$

$$= \frac{2}{3} u^{3/2} + 4u^{1/2} + c$$

$$= \frac{2}{3} \sqrt{(x-1)^3} + 4\sqrt{x-1} + c$$

(3)

b $\int_1^2 \frac{1}{e^x - 1} dx.$

$$= \int_1^2 \frac{e^x - e^x + 1}{e^x - 1} dx. \quad \checkmark$$

$$= \int_1^2 \frac{e^x - (e^x - 1)}{e^x - 1} dx.$$

$$= \int_1^2 \frac{e^x}{e^x - 1} - 1 dx. \quad \checkmark$$

$$= \left[\ln(e^x - 1) - x \right]_1^2 \quad \checkmark$$

$$= \ln(e^2 - 1) - 2 - \ln(e - 1) + 1$$

$$= \ln(e+1) - 1$$

(2)

Question 6

$$\underline{c} \quad \int_0^{\pi/2} \frac{1}{\cos x + 2} dx.$$

$$= \int_0^1 \frac{1}{\left[\frac{1-t^2}{1+t^2}\right] + 2} \cdot \frac{2dt}{(1+t^2)}$$

$$= \int_0^1 \frac{2dt}{1-t^2+2+2t^2}$$

$$= \int_0^1 \frac{2dt}{3+t^2}$$

$$= \left[\frac{2}{\sqrt{3}} \tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1 \quad (4)$$

$$= \frac{2}{\sqrt{3}} \cdot \frac{\pi}{6}$$

$$= \frac{\pi\sqrt{3}}{9}$$

$$t = \tan \frac{x}{2}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

$$= \frac{1}{2} (1 + \tan^2 \frac{x}{2}) dx$$

$$= \frac{1}{2} (1 + t^2) dx$$

$$\frac{2dt}{1+t^2} = dx$$

$$\text{when } x=0, t=0$$

$$\text{when } x=\frac{\pi}{2}, t=1$$

d i

$$I_n + I_{n-2} = \int_0^{\pi/4} \tan^n x dx + \int_0^{\pi/4} \tan^{n-2} x dx$$

$$= \int_0^{\pi/4} (\tan^n x - \tan^{n-2} x) dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x (\tan^2 x - 1) dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x \sec^2 x dx$$

$$= \int_0^1 u^{n-2} du$$

$$\begin{aligned} \text{let } u &= \tan x \\ du &= \sec^2 x dx \\ x &= \frac{\pi}{4}, u=1 \\ x &= 0, u=0 \end{aligned}$$

Question 6

d i continued.

$$\begin{aligned}
 &= \left[\frac{u^{n-1}}{n-1} \right]_0^1 \\
 &= \frac{1^{n-1}}{n-1} - \frac{0^{n-1}}{n-1} \quad | \quad (3) \\
 &= \frac{1}{n-1} - 0 \\
 &= \frac{1}{n-1}
 \end{aligned}$$

$$\text{II} \int_0^{\frac{\pi}{4}} \tan^5 x \, dx = I_5$$

$$= \frac{1}{4} - I_3$$

$$= \frac{1}{4} - \left(\frac{1}{2} - I_1 \right)$$

$$= -\frac{1}{4} + I_1 \quad |$$

$$= I_1 - \frac{1}{4}.$$

$$I_n \neq I_{n-2} = \frac{1}{n-1}$$

$$I_n = \frac{1}{n-1} - I_{n-2}$$

(3)

Now $I_1 = \int_0^{\frac{\pi}{4}} \tan x \, dx.$

$$= \int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos x} \, dx.$$

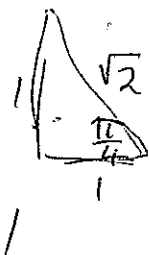
$$= \left[-\ln(\cos x) \right]_0^{\frac{\pi}{4}}$$

$$= \left[-\ln\left(\frac{1}{\sqrt{2}}\right) - -\ln 1 \right] \quad |$$

$$= -\ln\left(\frac{1}{\sqrt{2}}\right)$$

$$= \ln \sqrt{2}$$

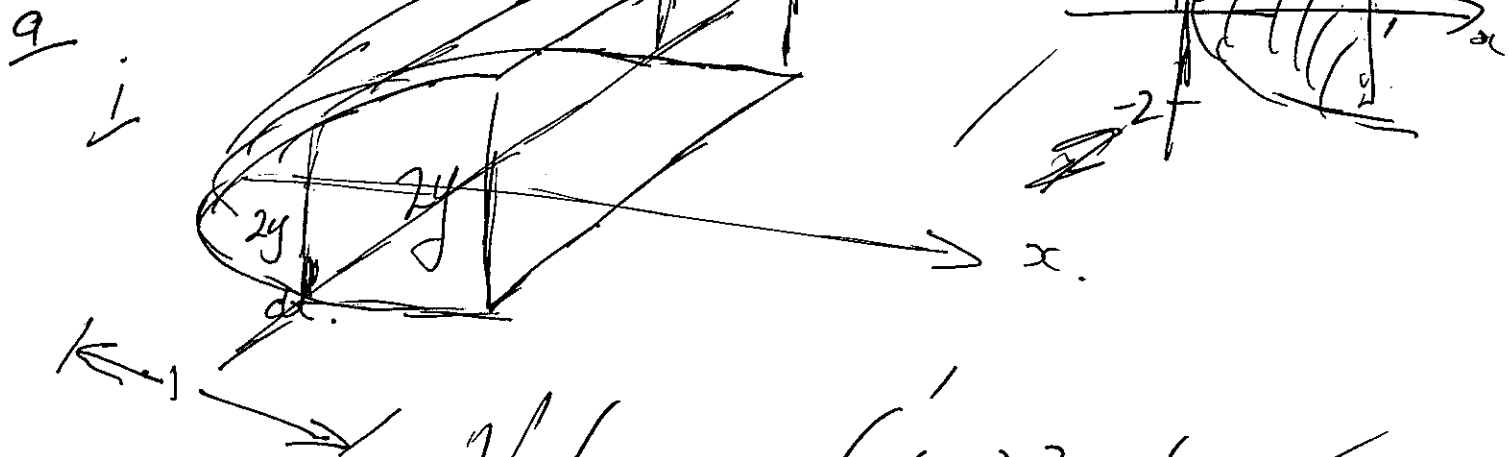
$$= \frac{1}{2} \ln 2.$$



$$= \frac{1}{2} \ln 2 - \frac{1}{4}.$$

$$= \frac{1}{4} (2 \ln 2 - 1).$$

Question 15) A solid is formed by revolving the curve $x = 1 - y^2$ about the y -axis.



$$\text{Volume} = \int_0^1 (2y)^2 dx$$

$$= 4 \int_0^1 y^2 dx$$

$$= 4 \int_0^1 4x dx$$

(4)

$$= 4 \left[2x^2 \right]_0^1$$

$$= 4 \times 2$$

$$= 8 \text{ units}^3$$

Question 7

b $x^3 + kx + 2 = 0$
 $x^3 + 0x^2 + kx + 2 = 0$

$$\alpha + \beta + \gamma = -\frac{b}{a}$$
$$= 0$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$
$$= k.$$

$$\alpha\beta\gamma = -\frac{d}{a}$$
$$= -2.$$

i $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma}$ (3)

$$= \frac{k}{-2}.$$

ii α, β and γ are the roots of $x^3 + kx + 2 = 0$

$\therefore \alpha^2, \beta^2$ and γ^2 are the roots of.

$$(\sqrt{x})^3 + k\sqrt{x} + 2 = 0.$$

$$\sqrt{x}^3 + k\sqrt{x} + 2 = 0.$$

(2)

iii If α a root of $x^3 + kx + 2 = 0$
then $\alpha^3 + k\alpha + 2 = 0$

that is $\alpha^3 = -k\alpha - 2$

Similarly $\beta^3 = -k\beta - 2$

and $\gamma^3 = -k\gamma - 2$

Question 7

b iii continued

adding gives

$$\alpha^3 + \beta^3 + \gamma^3 = -k\alpha - k\beta - k\gamma - 6$$

$$= -k(\alpha + \beta + \gamma) - 6 \quad \text{✓} \quad (2)$$

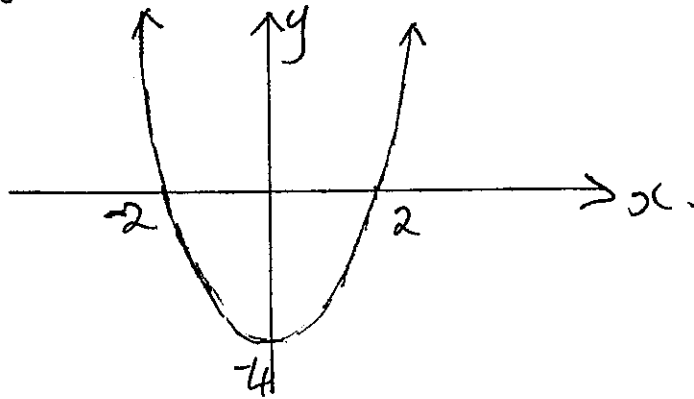
$$= -6 \quad \text{as } \alpha + \beta + \gamma = 0.$$

Question 8.

a iv $PS + PS' = e PM + e PM'$
 $= e (PM + PM')$
 $= e MM'$
 $= \frac{3}{5} \times \frac{50}{3}$
 $= 10$

(2)

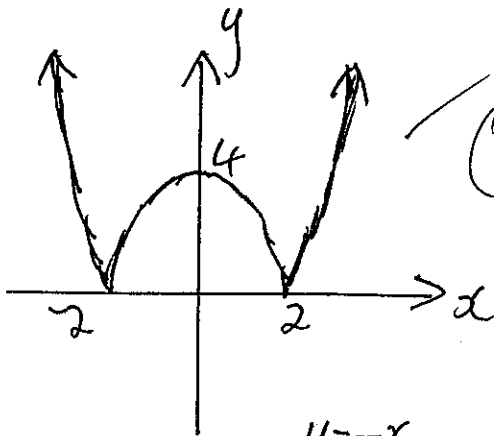
b i) $y = x^2 - 4 = (x-2)(x+2)$



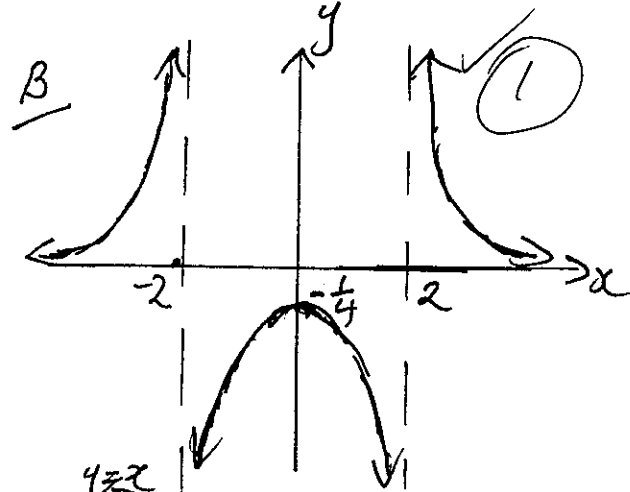
Ag (2)

(1)

ii
x

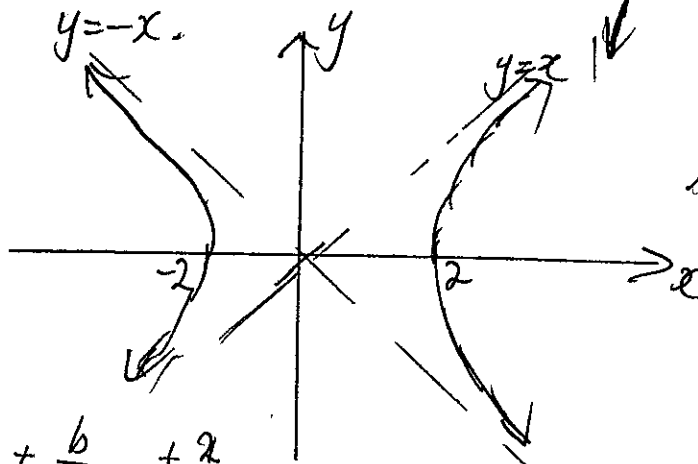


(1)



(1)

(1)



$$y^2 = x^2 - 4$$

$$x^2 - y^2 = 4$$

$$\frac{x^2}{4} - \frac{y^2}{4} = 1$$

Asymptote $y = \pm \frac{b}{a} = \pm \frac{2}{2}$

Question 8 (12)

i

$$\frac{x^2}{5^2} + \frac{y^2}{4^2} = 1.$$

$a=5, b=4 \quad b < a$

eccentricity, $e = \sqrt{1 - \frac{16}{25}}$

$$= \sqrt{\frac{9}{25}}$$

$$= \frac{3}{5}$$

(2)

ii Foci = $(\pm ae, 0) = (\pm 5(\frac{3}{5}), 0)$
 $= (\pm 3, 0).$

(1)

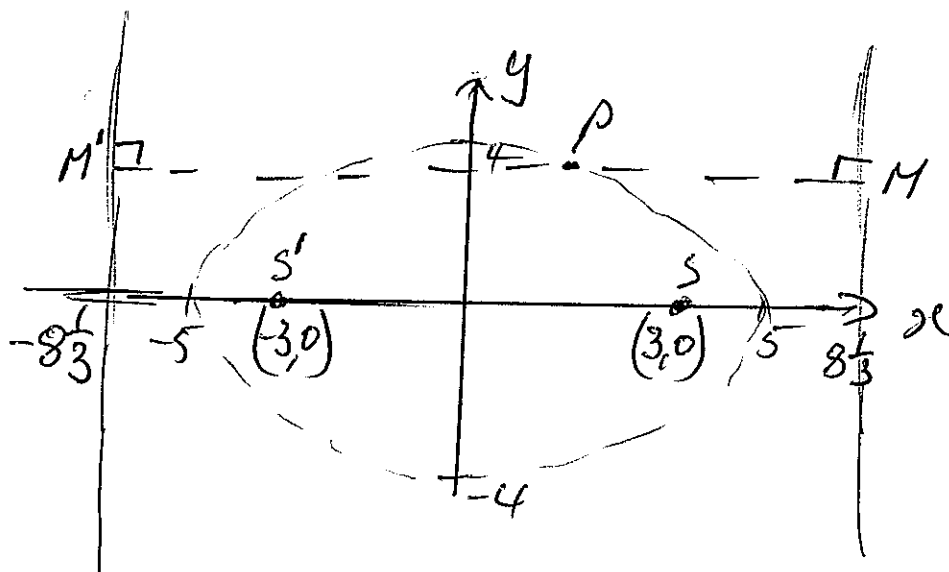
Directrices $x = \pm \frac{a}{e}$

$$= \pm \frac{5}{3/5}$$

$$= \pm \frac{25}{3}$$

$$= \pm 8\frac{1}{3}$$

(9/1)



(7/1)